1 **(a)** 32 to 34 **[B1]**

(b) Reading Upper Quartile as 40 and Lower Quartile as 23 to 24 **[M1]**

16 to 17.5 **[A1]**

(c) $120 - 100$ (or $120 -$ their reading at $t = 45$) **[M1]**

18 to 22 **[A1]**

[Total Marks: 5]

2 **(a)** **(i)** 120 **[B1]**

(ii) 108 (allow 107 to 109) **[B1]**

(iii) Upper quartile = 132 (allow 131 to 133) seen **[M1]**

24 (allow 22 to 26, or their UQ − their LQ) **[A1]**

(b) $80 - 68 = 12$ **[B1]**

[Total Marks: 5]

3 **(a)** **(i)** 24 **[B1]**

(ii) Reading Lower Quartile = 15 and Upper Quartile = 32.5 (Accept 32 to 33) **[M1]**

17.5 (Accept 17 to 18) **[A1]**

(b) Reading Cumulative Frequency at 45 km = 56 (Accept 55 to 57) **[M1]**

$60 - 56$ **[M1]**

4 (Accept 3 to 5) **[A1]**

[Total Marks: 6]

4 **(a)** **(i)** 50 **[B1]**

(ii) Lower quartile = 45 **[B1]**

Upper quartile = 56 **[B1]**

(iii) $56 - 45 = 11$ **[B1]**

(b) Reading cumulative frequency at $t = 60$ gives 70 **[M1]**

$80 - 70 = 10$ **[A1]**

(c) 15% of $80 = 12$ runners **[M1]**

Reading time at cumulative frequency 12 gives 40 **[A1]**

[Total Marks: 8]

5 **(a)** 6 points plotted correctly at the upper boundaries **[P2]**

(Allow P1 for 4 or 5 points plotted correctly)

A smooth curve drawn passing through all their points **[C1]**

(b) Reading from cumulative frequency 40 **[M1]**

$26 \leq$ answer ≤ 28 **[A1]**

[Total Marks: 5]

6 (a) 110 **[B1]**

(b) Upper quartile = 120 AND Lower quartile = 101 to 102 (reads CF at 30) **[M1]**

18 to 19 **[A1]**

(c) Reads cumulative frequency at 125 g as 99 to 101 **[M1]**

19 to 21 (from $120 - 100$) **[A1]**

[Total Marks: 5]

7 (a) 36, 72, 90 correctly entered in the table **[B1]**

(b) Plots $(10, 12)$, $(20, 36)$, $(30, 72)$, $(40, 90)$, $(50, 100)$ accurately **[P2]**

Allow [P1] if at least 3 points are plotted accurately, or if all 5 points are plotted at class midpoints

Smooth increasing curve drawn through all plotted points starting at $(0, 0)$ **[C1]**

(c) 34 (allow 33 to 35, or correct reading from their curve at cumulative frequency 80) **[B1]**

(d) Reads cumulative frequency at $t = 35$ as 82 (allow 80 to 84) **[M1]**

18 (allow 16 to 20, or $100 -$ their reading) **[A1]**

[Total Marks: 7]

8 (a) Values: $30, 64, 76, 80$ **[B2]**

Allow [B1] for 2 or 3 correct values.

(b) Points plotted correctly horizontally at upper class boundaries $(20, 40, 60, 80, 100)$ **[B1]**

Points plotted correctly vertically at heights $8, 30, 64, 76, 80$ **[B1]**

Smooth curve drawn through the points, anchored at $(0, 0)$ **[B1]**

(c) Reads curve at $t = 55$ correctly (expected value 53 to 57) **[M1]**

23 to 27 (Calculation: $80 -$ their CF value) **[A1]**

[Total Marks: 7]

9 (a) $90, 160, 188, 200$ **[B2]**

(B1 for 2 or 3 correct values)

(b) 5 points plotted correctly from their table **[B2]**

(B1 for 3 or 4 points plotted correctly)

Smooth curve drawn through their points **[B1]**

(c) Reading from curve at cumulative frequency = 140 **[M1]**

46 to 48 **[A1]**

[Total Marks: 7]

10 (a) Reading from cumulative frequency 60 **[M1]**

$22 \leq$ answer ≤ 24 **[A1]**

(b) Attempt to read from cumulative frequencies 90 (UQ) and 30 (LQ) **[M1]**

Lower quartile ≈ 15.5 and Upper quartile ≈ 29.5 seen **[M1]**

$12 \leq$ answer ≤ 16 **[A1]**

(c) Attempt to read at $d = 35$ (approx 105) **[M1]**

$120 -$ their reading **[M1]**

$13 \leq$ answer ≤ 17 **[A1]**

[Total Marks: 8]

11 (a) 46, 92, 112 **[B2]**

(Allow [B1] for 2 correct values)

(b) Points correctly positioned horizontally and vertically starting at $(50, 0)$ **[B1]**

All 5 points plotted correctly at class upper boundaries (Tolerance $\pm$ half small square) **[B1]**

Smooth curve drawn through all 6 points **[B1]**

(c) Finding target Cumulative Frequency = 60% of 120 = 72 **[M1]**

178 (Accept 176 to 180) **[A1]**

(d) Reading Cumulative Frequency at $m = 220$ is 100 **[M1]**

$120 - 100$ **[M1]**

20 (Accept 18 to 22) **[A1]**

[Total Marks: 10]

12 (a) All five missing values correct: $32, 76, 104, 116, 120$ **[B2]**

Award [B1] if at least 3 of the 5 values are correct

(b) All 6 upper-boundary points $(30, 10), (60, 32), (90, 76), (120, 104), (150, 116), (180, 120)$ plotted accurately as small crosses **[B2]**

Smooth increasing curve drawn through all plotted points (extending from $(0, 0)$ or $(30, 10)$ to $(180, 120)$) **[B1]**

(c) (i) Answer in the range 78 to 82 (reading at cumulative frequency 60) **[B1]**

(ii) $0.80 \times 120 = 96$ **[M1]**

Answer in the range 108 to 113 (reading at cumulative frequency 96) **[A1]**

(d) Reading cumulative frequency at $w = 115$ gives approximately 100 customers (accept 98 to 101) **[M1]**

Percentage $= \frac{100}{120} \times 100 = 83.3\%$ (or 85% of 120 = 102 customers) **[M1]**

No, the target is not met because $83.3\% < 85\%$ (or $100 < 102$) **[A1]**

[Total Marks: 11]

13 (a) Calculates 60% of 200 = 120 **[M1]**

54 to 56 **[A1]**

(b) Calculates 15% of 200 = 30, and targets CF of $200 - 30 = 170$ **[M1]**

70 **[A1]**

(c) Frequency for $0 < d \leq 30$ is 44 **[B1]**

Frequency for $30 < d \leq 50$ is 56 $(100 - 44)$ **[B1]**

Frequencies for $50 < d \leq 80$ is 88 AND $80 < d \leq 100$ is 12 **[B1]**

[Total Marks: 7]

14 (a) $0.8 \times 200 = 160$ **[M1]**

$28 \leq$ answer ≤ 29 **[A1]**

(b) Frequencies 30, 60, 80, 30 correct **[B2]**

(Allow B1 for 2 or 3 correct frequencies)

(c) Using midpoints 5, 15, 25, 35 **[M1]**

$\Sigma fx = 30 \times 5 + 60 \times 15 + 80 \times 25 + 30 \times 35$ (using their frequencies) **[M1]**

their $4100 \div 200$ **[M1]**

20.5 **[A1]**

[Total Marks: 8]

15 (a) Reading from curve at $m = 220$: 81 to 85 **[M1]**

100 − their reading **[M1]**

15 to 19 **[A1]**

(b) $p = 10$ **[B1]**

$q = 26$ **[B1]**

$r = 18$ **[B1]**

(c) Use of correct midpoints: $75, 125, 175, 225, 275$ **[M1]**

$\sum f \times m = 10 \times 75 + 26 \times 125 + 40 \times 175 + 18 \times 225 + 6 \times 275 = 16700$ **[M1]**

$16700 \div 100 = 167$ **[A1]**

[Total Marks: 9]

16 **(a)** Upper quartile = 66 (allow 65 to 67) and Lower quartile = 42 (allow 41 to 43) **[M1]**

24 (allow 22 to 26) **[A1]**

(b) Identifies 85th percentile: $0.85 \times 120 = 102$ **[M1]**

73 (allow 72 to 74) **[A1]**

(c) On average, the Physics scores are higher because the Physics median (62) is greater than the Mathematics median (55) **[B1]**

The Physics scores are more consistent (less spread out) because the Physics IQR (16) is smaller than the Mathematics IQR (24) **[B1]**

(d) Reads cumulative frequency at score 60 as 72 (allow 71 to 73), giving $120 - 72 = 48$ students scoring more than 60 **[B1]**

$\frac{48}{120} \times \frac{47}{119}$ **[M1]**

$\frac{94}{595}$ or 0.158 (allow 0.151 to 0.165) **[A1]**

[Total Marks: 9]

17 **(a)** Reading Lower Quartile = 38 (Accept 37 to 39) and Upper Quartile = 64 (Accept 63 to 66) **[M1]**

26 (Accept 24 to 29) **[A1]**

(b) Finding Cumulative Frequency = 15% of 200 = 30 **[M1]**

30 (Accept 29 to 31) **[A1]**

(c) Finding target Cumulative Frequency = $200 - 30 = 170$ **[M1]**

71 (Accept 70 to 72) **[A1]**

(d) Reading Cumulative Frequency at 45 marks = 74 (Accept 72 to 76) **[M1]**

Number of candidates NOT taking test = $200 - 74 = 126$ **[M1]**

$\frac{126}{200} \times 100\%$ **[M1]**

63 (Accept 62 to 64) **[A1]**

[Total Marks: 10]